PhD Project Plan

Project Plan — Monitoring and Predicting Droughts on National Scale: A Satellite-Integrated, Groundwater-Coupled Hybrid Modelling Framework for Denmark

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Abstract

1. PROBLEM STATEMENT AND MOTIVATION

The 2018 Danish drought caused agricultural losses exceeding DKK 6 billion and exposed a structural gap in the national early warning landscape. Operational systems such as **HIP** (Hydrologisk Informations- og Prognosesystem) and the **DK model** deliver high resolution diagnostics but lack probabilistic, satellite integrated outlooks at sub-seasonal to seasonal lead times. Meanwhile, macroscale land surface models such as Variable Infiltration Capacity (**VIC**), the model targeted by the DREAM call, lack a prognostic groundwater state. In a country where more than 99% of drinking water originates from aquifers, where water tables typically lie between 1 and 5 metres below surface, and where more than half of the agricultural area is sub-surface drained, this gap is not a detail but a first-order limitation. The recent finding by Forootan *et al.* (2024) that the severity of the 2017 to 2021 European drought is systematically underestimated in non-assimilated **W3RA** confirms that closing this loop, through groundwater coupling, multi-source data assimilation, and hybrid physics and machine learning integration, is both timely and necessary. The PhD will deliver a satellite-fed VIC Denmark system producing a probabilistic four-pillar combined drought index and short- to seasonal-scale predictions of water dynamics.

2. RESEARCH QUESTIONS

RQ1. Can a groundwater-coupled VIC Denmark, calibrated and constrained by joint assimilation of **GRACE FO** terrestrial water storage, **SMAP** and **ESA CCI** soil moisture, streamflow, and Jupiter piezometric heads, reproduce the propagation of meteorological, agricultural, hydrological, and groundwater drought signals at national scale better than uncoupled or non-assimilated baselines?

RQ2. Does a hybrid framework, combining differentiable parameter learning for VIC parameters with **LSTM** or Transformer post-processing trained on Catchment Attributes and Meteorology for Large-sample Studies, Denmark (**CAMELS DK**), recover predictive skill in groundwater-dominated Danish catchments, where pure LSTMs underperform (Liu *et al.* 2024)?

RQ3. Can a probabilistic four-pillar combined index (Standardised Precipitation Index (**SPI**)/Standardised Precipitation-Evapotranspiration Index (**SPEI**) plus Standardised Soil Moisture Index (**SSMI**) plus Standardised Streamflow Index (**SSI**) plus Standardised Groundwater Index (**SGI**)), aggregated via copulas with full ensemble and posterior uncertainty propagation, classify drought severity more skilfully than **EDO CDI v4**, and improve seasonal predictability when forced with bias-corrected Seasonal Forecasting System 5 (**SEAS5**) and Artificial Intelligence Forecasting System Ensemble (**AIFS ENS**) or GenCast?

3. APPROACH AND WORK PACKAGES

The project is organised into five interconnected work packages following a de-risked sequence.

- **WP1: VIC Denmark with groundwater coupling.** National VIC 5 at 1–10 km, parameterised from **VICGlobal** and refined with Danish hydrostratigraphy from **GEUS** and Jupiter wells. The groundwater module follows **SIMGM** (Niu *et al.* 2007), with a staged escalation path to 2D lateral coupling (Zhang and Liang 2021) if validation justifies it.
- **WP2: Joint calibration and multi-source data assimilation.** Local Ensemble Transform Kalman Filter (**LETKF**) and Constrained Bayesian (**ConBay**) **MCMC** assimilating **GRACE FO** terrestrial water storage (sub-monthly L1B, Retegui-Schiettekatte *et al.* 2025), **SMAP** and **ESA CCI** soil moisture (De Lannoy and Reichle 2016), streamflow, and Jupiter piezometric heads. Built on **PyGLDA** (Yang *et al.* 2025) with Fennoscandian Glacial Isostatic Adjustment (**GIA**) correction via ICE-6G_D.
- **WP3: Hybrid extension via differentiable hydrology.** Differentiable VIC trained via differentiable Parameter Learning (**dPL**, Tsai *et al.* 2021; Feng *et al.* 2022) on **CAMELS DK** (Liu *et al.* 2025), replacing per-basin calibration with gradient-based learning across 3,330 basins. An **LSTM** residual post-processor corrects biases in groundwater-dominated catchments (Liu *et al.* 2024).
- **WP4: Probabilistic four-pillar drought index.** Standardised sub-indices spanning the full propagation chain — **SPI**/**SPEI**, **SSMI**, **SSI**, **SGI** — aggregated via copulas into a probabilistic Multi-pillar Combined Drought Index (**MCDI**) with full posterior probabilities of drought severity categories.
- **WP5: Short- to seasonal-scale forecasting.** VIC DK forced with **SEAS5** and **AIFS ENS** or GenCast (Price *et al.* 2025) for 1–6 month outlooks, evaluated via Continuous Ranked Probability Score (**CRPS**), Brier score, and ROC, with reverse Ensemble Streamflow Prediction (**ESP**) diagnostics. Benchmarks: **EFAS Seasonal** and **EDgE**.

4. EXPECTED CHALLENGES AND MITIGATIONS

Four principal challenges are anticipated, each with a concrete mitigation strategy.

- **Scale mismatch.** GRACE FO has a native resolution of approximately 300 km versus Denmark's 43,000 km^2. Mitigations: sub-monthly L1B assimilation (Retegui-Schiettekatte *et al.* 2025), **ConBay** merging with SMAP (Mehrnegar *et al.* 2023), ML-based downscaling against Jupiter wells, and **GIA** correction via ICE-6G_D.
- **Equifinality.** Multiple parameter sets can reproduce equivalent streamflow but incorrect partitioning between evapotranspiration, soil moisture, and recharge. Mitigations: GEUS-derived priors on aquifer parameters, localisation and inflation within **LETKF**, and **dPL** as deterministic precursor before stochastic assimilation.
- **ML physical consistency.** Pure LSTMs may extrapolate poorly and do not guarantee mass conservation. Mitigations: keeping VIC as the physical backbone, restricting the LSTM to a residual post-processor role, and applying conformal prediction for honest uncertainty intervals.

- **Index collapse.** Naive averaging of indices with different time scales loses causal propagation and tail dependence of extremes. Mitigations: pillar-appropriate scales (SPI 1–3 months, SGI 6–12 months), copula aggregation preserving tail dependence (Hao and AghaKouchak 2013), and validation against Naturskaderådet claims, crop yield anomalies, and EDO CDI v4.

5. FEASIBILITY, FIT AND OUTPUTS

The plan is feasible within three years because Year 1 already delivers a publishable physics-based baseline, even if the hybrid extensions slip. The **AAU PyGLDA**, **ConBay**, and **HPC Claaudia** infrastructure provide a strong starting point, while **CAMELS DK**, Jupiter, **DMI Klimagrid**, and **HIPRAD** supply the data backbone. The project aligns with the Villum Young Investigator Programme on "Synergy of Physics-based and Data-driven Methods" and complements **GEUS DK model** expertise, opening natural collaboration with Henriksen, Stisen, and Schneider.

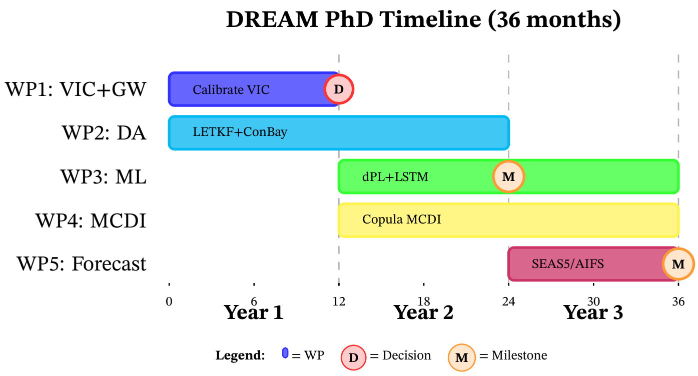

Figure 1. DREAM PhD timeline showing five interconnected work packages over 36 months. **WP1** establishes a physics-based VIC-Denmark baseline with groundwater coupling (Year 1), followed by a critical decision (D) on 2D lateral coupling. **WP2** (LETKF + ConBay data assimilation) runs in parallel throughout, constraining all physical parameters. **WP3** (differentiable hydrology + LSTM hybrid) and **WP4** (copula-aggregated multi-pillar drought index) execute in Years 2–3 in parallel. **WP5** (SEAS5/AIFS seasonal forecasting) focuses on Year 3 outputs. The timeline demonstrates de-risking: Year 1 alone delivers a publishable physics baseline even if hybrid extensions slip. Milestones (M) mark completion of hybrid competence (month 24) and operational prototype readiness (month 36).

Expected outputs include:

- Three first-author papers in *Water Resources Research*, *Hydrology and Earth System Sciences*, and *Remote Sensing of Environment* or *Science of The Total Environment*
- An open-source **VIC DK** plus **ConBay** data assimilation codebase released through PyGLDA
- An operational **MCDI** prototype co-designed with **DMI**, **GEUS**, and Naturskaderådet end-users

6. BACKGROUND AND ADDED VALUE

My MSc thesis at Politecnico di Milano, where I built, calibrated, and ran sensitivity and uncertainty analysis on the **FEST** process-based hydrological model over the Po basin, gave me direct experience of the central methodological tension that this project addresses: constraining subsurface parameters (hydraulic conductivity, deep percolation factors, **SCS** curve number) from surface observations alone.

My current operational role at Progesi S.p.A., where I run multi-hazard monitoring with **ICON** and **ECMWF IFS** forcing, satellite-derived diagnostics, and Python, R, and MATLAB workflows for Regione Lombardia, translates directly to the operational demands of a national-scale drought early warning system.

The combination of process-based modelling, applied data science, and recent exposure to operational hydrometeorology positions me to start contributing to WP1 and WP2 immediately, while developing the hybrid machine learning competence required for WP3 through structured upskilling in **NeuralHydrology**, PyTorch, and differentiable hydrology.

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